EBLEX BEEF BRP MANUAL 7

Better Returns Programme

Feeding growing and finishing cattle for Better Returns











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Contents











Page 2 **Growing and finishing phases**

Page 3 **Feeding for growth**

Page 4 **Maximising dry matter intake**

Page 5 **Ration guidelines**

Page 6 **Making the most of grass**

Page 7 **Growing housed cattle**

Page 8 **Growing to finishing the big change**

Page 9 **Finishing housed cattle**

Page 10 **Feed conversion efficiency**

Page 11 **Feeding for carcase quality**

Page 12 **Optimising nutrient balance**

Page 13 **Choosing the supplement**

Page 14 **Supplementing silage and straw**

Page 15 **What is the dung telling you**

Page 16 **Top 20 energy feeds**

Page 17 **Top 10 protein feeds**

Growing and finishing cattle requires attention to detail at all stages to be able to match market needs and make the best return on your investment. Best returns comes from cattle that are well grown and efficiently finished. Feeding not only determines animal performance but also contributes to animal health, cost of production and therefore profit margins.

This BRP manual will illustrate the importance of distinguishing between the rearing, growing and finishing phase. As well as formulating rations according to animal requirements to optimise animal performance appropriate to the system and meet market specifications.

The role of feeding in determining carcase quality is discussed, along with advice about choosing the best feeds for your cattle to meet their nutrient requirements.

The manual complements our earlier publication BRP Mini Feeds Directory which provides guidance from conception to market on using effective nutrition to achieve better returns on a range of feeds.



John Cross

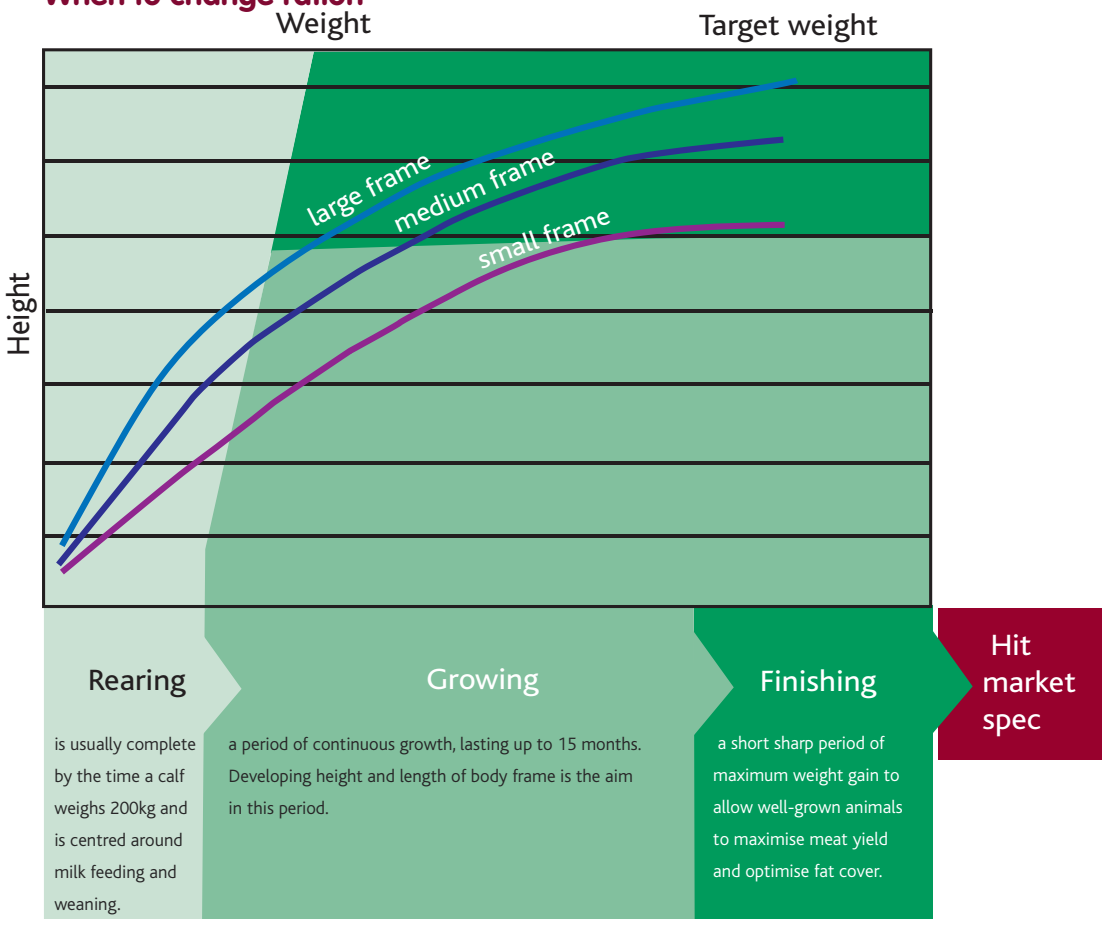
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Growing and finishing phases

Feeding management of beef cattle divides into three distinct phases. Each requires different ration specifications.

When to change ration



The length of each phase varies with breed, frame size and target market. Native breeds or small framed cattle usually require a longer growing, but shorter finishing period than larger continental cattle.

Feeding management is crucial to meet market specification

Know your end market; cattle finish at a variety of weights, different ages. Target specifications will vary depending on breeding and the target market. Here are some examples.

Main market	Target age	Type	Carcase weight	Classification
Butchers	16-24 months	Heifers, steers	250-320kg	R4L or better
Supermarket	16-30 months	Heifers, steers	280-380kg	R3 or better
Manufacturing beef	12-30 months	Bulls, heifers, steers	280-400kg	-02 or better

Please note there will be variations on the above examples.

Feeding to grow

Understanding your stock and system will help you feed to exploit growth potential

Your choice of beef production system should be based on your available resources.

Stock – is determined by:
Gender – bull, steer or heifers
Mature size – small, medium or large framed
Type – Native or continental.

System – is determined by:
feed, land, buildings, machinery, labour, finance.

The frame size of an animal is a useful guide to its growth potential and which feeding system it is best suited to. Frame size determines mature liveweight and is affected by breed type and breeding within a breed.

When assessing frame size in growing cattle take account of breed, age, weight and height.

Large-framed animals, tall for their age, suit markets which target fast daily liveweight gain of over 1kg/head/day.

Short–framed animals, short for their age, suit markets with slower target daily gain– under 0.8kg/head/day.

Medium-framed animals of average height for their age are best grown between 0.8 and 1kg/head/day and can be targeted to a number of markets.

Pointers Use the rails in a race to give a guide to frame size.

Animal	Target height gain/month Small frame	Medium frame	Large frame
200kg – 300kg	3cm	4cm	5cm
300kg – 400kg	2cm	3cm	4cm
400+kg	1cm	2cm	3cm



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Maximising Dry Matter Intake

Maximising dry matter intake is vital to better growing and finishing of cattle. Improving animal performance reduces days to slaughter and cuts production costs.

It is important to present fresh palatable feed and have water available at all times to optimise intake and remember the old saying a dry bed is as good as a feed.



A well-designed and managed feed area can be as important as what is fed.	
✓	very smooth, clean surface
✓	eating surface about 10cm above hoof height
✓	brisket board, preventing food being pulled under foot, suited to height of cattle
✓	clean out refused food at least 3 times a week
✓	site water troughs to avoid feed being splashed
✗	pitted concrete that can be rough on the tongue
✗	rub marks from neck rail – shows wrong position
✗	stale feed build up

Ration guidelines

Growing animals have a large appetite compared to body weight. Thus they thrive on high levels of medium energy forage, as long as there is enough degradable protein to 'fuel' microbial activity in the rumen.

Growing cattle – controlled continuous frame growth
DMI = 2.3% bodyweight
0.7–1.3kg/day LWG
30–60% DM (dry matter)
*15–16% CP (crude protein)
10.5–11.4 MJ/kg DM ME (metabolisable energy)
*>40% NDF (neutral detergent fibre) – digestible
*<3% fat
<20% starch and sugar
*0.8% calcium

Finishing cattle – short sharp period of maximum LWG
DMI = 2% Bodyweight
>1.4kg/day LWG
30–60% DM (dry matter)
*12.–15% CP (crude protein)
>12.2 MJ/kg DM ME (metabolisable energy)
*>25% NDF (neutral detergent fibre) – digestible
*6–8 % long fibre
*<6% fat
>33% starch and sugar
*0.6% calcium

Growing rations should be high in both structural fibre eg straw and digestive fibre eg sugar beet pulp, high in protein, high in minerals and vitamins and with a moderate energy density. High levels of starch are not recommended in this period as this can lead to unwanted fat deposition, especially in small-medium framed, 'easy fleshed' cattle. Some 'easy fleshed' growing cattle can stay on this ration till finish to avoid getting over fat while underweight.

Finishing cattle are often heavier than growing cattle, with a reduced appetite in relation to body weight. Rapid liveweight gain requires feeding high energy cereal based feeds. High starch levels can promote fast weight gains and more efficient feed conversion particularly in the larger framed, more difficult to flesh animals.

The key to profitable finishing is feed conversion efficiency (FCE), obtained by maintaining high dry matter intakes and live weight gains. Adequate quantities of structural fibre (0.5– 1.5kg straw/head/day) is essential.



Animals that spend long periods lying down chewing the cud, are an indication of efficient feed utilisation and good rumen health.

* expressed relative to DM

Making the most of grass

Grazing

Grazed grass is potentially the cheapest feed available. To achieve the best potential growth consider: stocking rates, sward, grazing quality and quantity.

The use of modern grass varieties along with appropriate types of clover in the sward can bring big yield benefits as well as reducing nitrogen fertiliser requirements. Target gains at grass depend on type of stock, available grazing type and length of grazing season. Large-framed cattle turned out at 250-300kg onto good leys with a long grazing season can gain 200kg. Small-framed cattle on marginal grazing may gain just half this weight.

When grazing is insufficient supplementation will be needed:

Concentrate supplementation	High quality forage-based buffer ration
• low levels maintain contact with stock	• maintains growth rates
• higher levels help maintain growth rates	• reduces grazing time
• does not reduce grazing time	• allows grass regrowth

Finishing cattle on grass

High levels of grass quality and quantity must be maintained for as long as possible to take 'easy fleshed' cattle through to finish. Bigger framed cattle struggle to achieve a good finish on pasture alone in summer and autumn. Finishers can be kept out on full ad-lib finishing feed as long as ground is dry enough to avoid poaching and there is a dry area to lie down.

Keep lot sizes small and move bunkers, troughs and feed trailers regularly to avoid poaching .

Silage

For growing cattle grass should not be allowed to mature beyond 50% ear emergence at cutting. Feeding silage with straw delays progress through the rumen allowing improved digestion.

For finishing cattle, high quality grass silage can be a digestible fibre source replacing sugar beet pulp, maize gluten, etc. However, fed at high levels alongside high starch feeds can result in loose dung.

Alternatively, silage can be made very mature with a long chop length. Fed in small quantities it offers a structural fibre source to keep the rumen healthy when high levels of concentrates are fed.



Growing housed cattle

Grower rations are mainly forage-based (silage, whole crop, straw and combinations of these). These can be supplemented at least once a day with dry or moist feeds to add energy, protein, minerals and vitamins. Rations can be formulated with a wide range of home-grown and bought-in feeds. The most likely scenario for a winter ration would be grass silage plus a blend. It is important that silage is palatable and consistent quality, it also helps if it contains reasonable levels of protein. Generally, only wholecrop legume silages can be fed without protein supplementation. The formulation and quantity of supplement depends on silage analysis and intake. Adding straw slows down the rate of passage of silage and promotes cud chewing.

Feeding for growing – starting the process

Cattle already on the farm

Introduce suckler calves to creep feeding as and when required depending on target growth rates, available milk and grazing. Begin at least 4 weeks before weaning to minimise 'weaning check'. Creep feed should be palatable, dust-free and contain high levels of digestible fibre.

Creep feed can include trace elements and vitamins to boost immune status before respiratory disease challenges in early winter.

Typical creep ration:
14–16% crude protein in DM
11–12 MJ/kg DM ME
35% NDF (neutral detergent fibre) - digestible
Target gain: 0.8 – 1.3kg/day LWG

Bought-in cattle

Cattle, tired after a journey, should be kept away from other stock in a draft-free, well bedded pen, with plenty of space, good access to palatable forage-based feed with long fibre and clean water.

After 12-18 hours rest:

- check ear tags and weigh each animal
- health check in accordance with farm's health plan
- if necessary, treat and isolate
- vaccinate.

Then group animals by weight and type. Gradually introduce them to their new ration before moving to new accommodation.



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Growing to finishing – the big change

Timing is the key to changing successfully from the growing to the finishing ration.

Targets	Days on finishing ration	DLWG	Total LW gain
Large framed	80–100	1.5kg	120-150kg
Medium framed	60 – 80	1.4kg	85kg
Small framed	0–60	1.3kg	0-50kg

Finishing rations contain less forage and more supplement than growing rations which leads to increased rumen aciditiy which challenges animal health. The rumen microbial population has to change from supporting a large population of fibre digesting bacteria to supporting a large population of starch digesting bacteria.

This costs time and money, so helping the animal to adapt quickly delivers efficiency. The finishing ration should be introduced gradually to the growing ration so that the high energy feeds are fed in increasing amounts each day over a period of a week and the amount of forage available is reduced at the same time. Alternatively, if the cattle are being introduced to ad lib feeding from a hopper then the concentrate or blend should be fed at gradually increasing amounts over a week until the cattle are not eating all the ration between feeds.

Bulls – changing directly from rearing to finishing

Bulls around 200kg live weight can move from the rearing ration straight to the finishing ration. However they can be on the finishing ration for 10 months and can become hard to finish. It is best to introduce a slightly higher energy and more palatable '2nd stage finisher' ration for the last 60 to 80 days to give them a boost. Suckler bred or beef cross bulls from the dairy herd weighing over 200kg liveweight will respond best to finishing rations formulated with protein levels of approximately 15% crude protein (CP). Whereas, pure dairy bulls over 200kg liveweight will respond better to lower protein levels of approximately 12% CP. Feeding these bulls on high protein diets tends to make them grow too tall and lean, making them very hard to finish.



Introduce finishing ration over seven days to adapt rumen bacteria

Finishing housed cattle

Finishing cattle – short sharp period of maximum gain

During finishing appetite falls in relation to bodyweight so intake should be encouraged through feed access, freshness and palatability.

- Drier rations stay fresh longer in the trough.
- Mix dry and wet rations to avoid very wet or very dry rations.
- Protein requirement should be met by microbial protein, but rumen microbes need a small amount of degradable protein. If insufficient rumination takes place adding a small amount of degradable protein will help. If the best cost option feeds available are high in protein the CP level can rise to a maximum of 17% before performance is compromised.
- Energy, particularly from starch, is all important to drive fermentation in the rumen. But at very high energy densities (over 13 MJ ME/kg DM) intake will be compromised.
- A small amount of long, hollow tube fibre, eg straw, which floats and scratches the rumen walls promotes rumination, prevents acidosis and improves feed conversion efficiency.
- Too much fat and oil coats fibre particles in the rumen and prevents breakdown by rumen microbes, leading to acidosis and reduced intake. Brewers grains, oil-coated chips and biscuit meal are good feeds but have upper inclusion limits. If two or more are fed together keep well below the upper limit.
- Requirements for major minerals are reduced but trace elements and vitamins are still important.

Bull beef

Keeping male calves entire only helps FCE if they are kept growing at maximum gain. In a store period bulls have no better feed conversion than steers.

Grass silage has no major role in bull beef diets. Small amounts help meet digestible fibre requirements, but large amounts will reduce energy density and extend days to finish.

- Minimise aggressive behaviour to promote profitability.
- Keep group sizes small – no more than 20.
- Avoid mixing batches which leads to fighting and riding.
- Keep bulls away from other stock, particularly breeding stock.
- Keep bulls away from noisy farm activity.
- Feed and bed in the morning and leave them in peace in late afternoon and early evening. In the evening, as the herd instinctively settles down for the night, if left undisturbed head butting and pushing occurs in an assertive, but non-aggressive way. If this does not happen stress and aggression will result in reduced performance.



Improving feed conversion efficiency (FCE)

Genetic and environmental factors influence FCE. On the same ration, bulls are more efficient than steers which are more efficient than heifers. Within a gender, breeding plays a large part in determining FCE.

Improving the environment (including feeding) can be achieved far faster than genetic change.

Areas to target:

Animal bodyweight-

- FCE reduces as animals get older and heavier.

Animal health

- Worm burden soaks up inputs and depresses performance.
- Laminitis reduces feeding time and intake.
- Disease – even sub-clinical disorders can reduce performance.

Stress

- Aim for group sizes of 20 at most.
- Ensure every animal has space to lie down on dry bedding.
- Avoid mixed sizes to avoid bullying.
- Ensure feed and water are fresh.
- Ensure good air quality – increased breathing rates reduces weight gain.

Nutrition

- Structural fibre eg straw promotes cud chewing and lowers rumen acid load. Correct length is vital – just short enough to avoid rejection but no shorter than the muzzle width of the animal.
- Balanced energy sources mainly from starch with some digestible fibre eg sugar beet pulp increases weight gain.
- Protein balanced in speed of degradability to suit the energy sources will increase rumen microbial population and weight gain.
- Minerals and vitamins should be balanced to rest of diet.

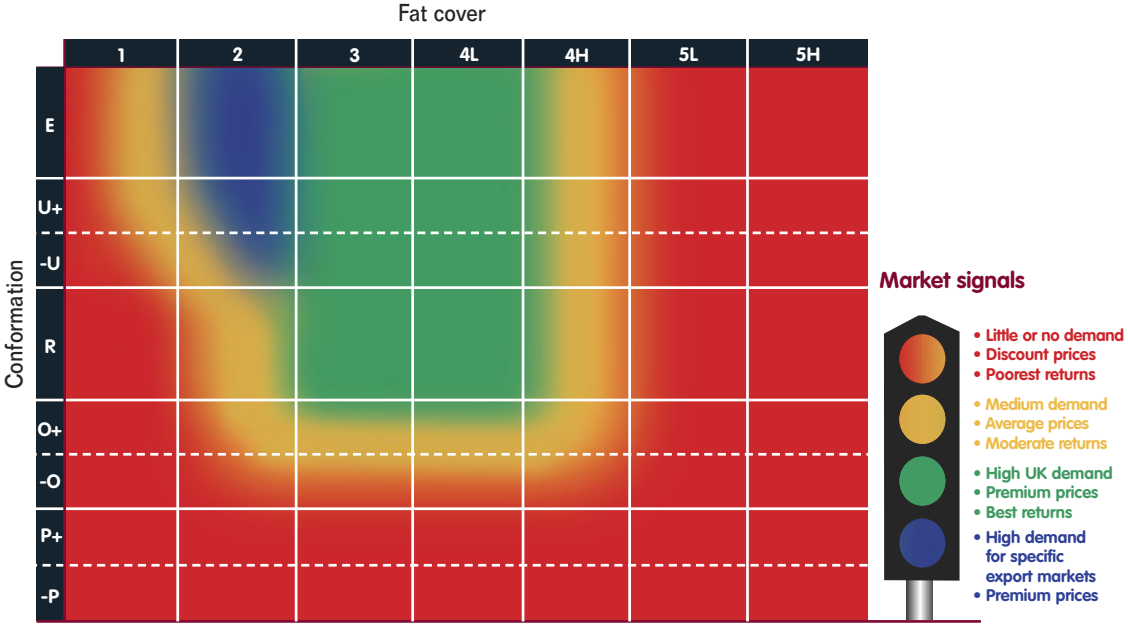
Feed

- Improving FCE reduces feed cost/kg gain.
- Farm-grown feed can cut cost and often boost performance.
- Selling high starch cereals and buying lower starch cereal by-products can reduce cost effectiveness due to haulage costs and potentially lower performance.

Animal weight	Dry Matter appetite	% of daily feed intake maintenance	Likely DLWG from balance of feed	FCE	Cost/kg LWG (£145/ton DM)
450kg	10kg	42%	1.2kg	8.3:1	£1.20
550kg	11kg	44%	1.15kg	9.6:1	£1.39
650kg	11.8kg	46%	1.1kg	10.7:1	£1.55
750kg	12.5kg	48%	1kg	12.5:1	£1.81

At these costs, carcase weights over 380kg or liveweights over 650kg are hard to justify as feed cost/kg gained exceeds value/kg gained. Depending on outlet, taking bulls to these heavy carcases may be viable provided liveweight gains exceed 1.6kg/head/day.

Feeding for carcase quality



Fast finishing maximises muscle development and improves conformation. Animals that are finished slowly (1.2kg/day) grading R could have been a -U if finished faster (1.5kg/day). However faster finishing cannot turn an -0 into a U+ that relies on breeding.

Faster finishing will take animals through fat classes faster. Animals finished at1.2 kg/day LWG can take six weeks to go from 4L to 4H, whereas animals finishing at 1.5+kg/day can take just three weeks. Therefore, it is important to select the animal for market at the right stage.

Fat colour

Some markets require carcasses with white fat (mainly bulls for export). Fat colour comes from fat soluble pigments in plants. These range from very strong pigments in carrots, fairly strong pigments in lucerne and normal pigments in grass. Some pigment colour is lost in the silage making process so grass silage will not colour fat as much as grazed grass. Also pigments accumulate in fat over time so with older animals it is harder to achieve white fat.

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Optimising nutrient balance

The most important macro nutrients to consider for beef cattle nutrition are energy (mainly from carbohydrates) and protein. These nutrients can be present in feeds in different forms that make them available to the animal at different rates. A balanced release of soluble carbohydrate and rumen degradable protein is required to fuel fermentation in the rumen and optimise cattle performance.

Carbohydrate fermentation

Fibres	Starch	Sugars
Straw	Cracked cereals	Molasses
Low D silage	Crimped cereals	
Soya hulls	Biscuit blends	
Palm kernel	Potatoes	
Citrus pulp	Rolled cereals	
High D silage	Ground cereals	
Sugar beet pulp		

slow → **fast**



Protein degradation

Digestible undegradable (DUP)		Effective rumen degradable protein (ERDP)
Prairie meal (maize gluten feed)	Distillers grains	Rapeseed meal
Soyabean meal	Beans	Pot ale syrup
Lupins	Peas	Feed grade urea

Details and rationing advice for a range of feeds can be found in the *BRP Mini Feeds Directory*.

Also consider fibre:

As well as providing a source of energy in ruminant diets fibre is required for efficient rumen function.

Fibre has two forms, either:

Structural fibre – supplied by long stemmed, mature forages eg straw, hay. Chop length of this material should be 5–10cm long, suited to the muzzle width of the animal your are feeding.

Digestible fibre – supplied by products such as sugar beet pulp and high D-value silage.

The content of digestible fibre in the total ration DM should be between 25–40%.

It is important to appreciate the difference between these two fibre sources, eg sugar beet has a high digestible fibre content but provides little structural fibre, whilst straw has a low digestible fibre content but is a valuable source of structural fibre if not chopped to short.

Choosing the right supplement for your ration

Rapidly fermentable carbohydrates in the form of molasses and finely ground cereals should be fed with rapidly available nitrogen from urea or rapeseed meal.

Slower forms of fermentable carbohydrates such as high D-value grass silage or sugar beet pulp should be fed with slower forms of degradable protein such as soyabean meal.

Very indigestible forms of fibre such as low D-value hay, silage or cereal straw need to have some rapidly fermentable carbohydrate and rapidly available nitrogen fed along side them to 'kick start' the rumen fermentation. Too much rapidly fermentable material and the rumen empties too quickly for the fibre to break down and large quantities of undigested fibre will come through in the faeces. Not enough and the fibre stays in the rumen for hours suppressing the animals appetite and holding back its performance.

Comparing supplements

When choosing supplements consider:

- Nutrient composition
- Quality and consistency
- Storage needs
- Medium term availability
- Complementarity with other feeds in ration

Feeds should be compared at the very least on the basis of £/tonne DM, taking into account their relative feed values. Comparisons made on the basis of cost per unit metabolisable energy or crude protein (CP) enable feeds to be compared on the basis of the nutrients they supply to the diet.



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Supplementing silage and straw growing diets

Example: A 300kg animal eating 7kg DM with a target daily gain of 1kg.

Both rations below supply the same amount of energy and protein but it is important to notice that since less supplementation is required with the better quality silage, the protein concentration in the supplement is greater. Poorer quality silage is wetter and less is eaten on a dry matter basis hence more supplement is required to meet energy requirements.

Silage based rations	kg FW	kg DM
Straw	0.5	0.4
Good silage 29% DM, 14% CP, 11 MJ ME/kg DM	20.0	5.7
Blend 25% CP	1.0	0.9

Straw	0.5	0.4
Poorer silage *21% DM, 12%CP, 10.2 MJ ME/kg DM	20.0	4.2
Blend 20% CP	2.8	2.4

Straw has been included to add scratch factor and improve rumen function

Other options for growing rations can include straw as the base forage.

Straw based rations	kg FW	kg DM
Straw	2.5	2.15
Blend 18% CP	5.5	4.8

Straw	2.0	1.7
* Sugar beet pulp	2.5	2.3
* Brewers grains	12.5	3.1

* 5 parts grains to 1 part sugar beet pulp

Ration 11 MJ ME/kg DM, 15% CP in DM

What is your dung telling you?



Looking at the dung is a useful indicator of how well the rumen is functioning and how well an animal is digesting their diet.

What the manure looks like	Possible dietary causes (not disease)
Loose manure	• High protein (total or soluble); pasture • Insufficient fibre in ration
Diarrhoea	• Spoiled, mouldy feed or silage • Ruminal acidosis • Increased hindgut fermentation
Foamy manure or mucin casts (pieces of gut lining that indicate gut damage)	• Ruminal acidosis • Increased hindgut fermentation
Large particles, undigested feed	• Cattle are sorting feed; slow feeding • Not enough chewable fibre/forage/feed • Ruminal acidosis
Manure variable in a feeding group	• Cattle are sorting feed; feeding slowly • Spoiled, mouldy feed or silage (spoilage does not usually mix uniformly)

Source: US Dairy Forage Research Center, Wisconsin

If the rumen is working well, you should see:

- few feed particles over 1.5cm long
- little recognisable feed
- 80% of cattle not sleeping, eating or drinking should be ruminating.

Looking at the dung should be used alongside other observations/measurements to evaluate successful ration formulation:

Cattle	Performance, coat, lameness, rumination
Feed	Availability, mould/dust, analysis, consistency, mixing, refusals
Clamp/feed storage	Mould, clean, fresh, heating, mixing
Water	Clean, fresh, available
Housing	Comfortable, used, clean and ventilated
Employees	How they treat the cattle

Top 20 energy feeds

There are a variety of feeds available to feed to finishing cattle, some are more suitable than others. This chart shows the most popular 20 energy feeds listed by their effectiveness in producing high levels of performance in terms of weight gain, feed conversion efficiency and flexibility of use with other feeds.

		MJ ME/kg DM	Starch %*	Sugar %*	Strengths	Weaknesses
1	Maize grain	14.3	71	2	Slow degrading starch	Low protein
2	Wheat	13.8	69	3.5	High starch	Low fibre
3	Bread	14	69	4.7	High starch	Low fibre
4	Triticale	13.4	66.5	4	High starch	Low fibre
5	Potatoes	13.5	62	8	High starch	Low fibre
6	Barley	13.2	59	3	High starch	Low fibre
7	Biscuit blends	15	49	9	High starch	Low fibre
8	Breakfast cereal blends	14	52	6	High starch	Low fibre
9	Sugar beet	12.3	3	69	High sugar	
10	Fodder beet	12.1	2	65	High sugar	
11	Good maize silage	11.8	35	2	Low cost	Can be variable
12	Molasses	12.6	–	65	High sugar	Low fibre
13	Beet pulp	12.5	1	20	High fibre	Low starch
14	Maize gluten	12.5	21	3	High fibre	
15	Oats	12.2	42	1	High fibre	
16	Good cereal whole crop	11	25	3.3	Low cost	Can be variable
17	Wheat feed	11.5	27.5	7	High fibre	Low energy
18	Citrus pulp	12.5	6.5	24	High fibre	Low starch
19	Soya hulls	11.9	5	3	High fibre	Low starch
20	Palm kernel	12.3	4	3	High fibre	Low starch

* expressed relative to DM

Top 10 protein feeds

This table shows the most popular 10 high protein feeds listed by their effectiveness in producing high levels of performance in terms of weight gain, feed conversion efficiency and flexibility of use with other feeds.

		MJ ME/kg DM	Starch and sugar %*	Crude protein %*	Strengths	Weaknesses
1	Maize distillers	14.8	5.5	28	High energy	–
2	Wheat distillers	13.5	8.6	32	High energy	–
3	Soya	13.8	14	52	High energy	–
4	Lupins	14.3	13	38	High energy	–
5	Pot ale syrup	14	3	37	High energy	–
6	Beans	13.8	47	29	High starch	Low fibre
7	Peas	12.8	53	24	High starch	Low fibre
8	Linseed	13	11	37	–	Low fibre
9	Rapeseed meal	12.1	14.5	38.5	High fibre	Lower energy
10	Urea	0	0	287	Very low inclusion	Must be well mixed

* expressed relative to DM

Vitamin and mineral supplementation

Vitamin and mineral supplementation should be included in all rations according to the requirements of the class of stock being fed and the composition of the ration. Deficiencies will lead to reduced performance.

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